

SURVEYS FOR “EXPLANATIONS”

Thomas Graeber

Christopher Roth

Constantin Schesch

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1 Orator experiment

Welcome

In this survey, we will ask you to record voice messages to answer different questions. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.).

Please make sure you are in a quiet environment. You will only receive your completion payment if your voice is clearly recorded.

You are eligible for a bonus of \$10 for one of your recordings! The details of how you might receive a bonus will be explained on the following pages.

For each question, your recording will actually be played to another respondent. The other respondent will have to answer the same question as you after listening to your recording.

Privacy & Anonymity

All voice messages are treated strictly anonymously. They will never be linked to your person and will never be published anywhere. This data will be used solely for academic research. You can therefore talk freely and informally in each voice message. The other respondent who will listen to your voice recording will not be given any more information about you: your anonymity will be preserved.

Test Your Microphone

Use the recorder below to test your microphone. Click “Record”, say the sentence “The dog runs in the park.”, then click “Stop Recording & Submit”. You may have to give your browser permission to access the microphone after you click “Record”. After a recording, it might take the website a few seconds to upload your recording: please be patient.

Trial Question

This trial question is an attention check. To be eligible to participate in this study, we simply ask you to record a voice message that lasts at least 20 seconds on your thoughts about this topic. There is no correct answer, and it does not matter what you say, as long as you record a message that lasts at least 20 seconds.

On how many days in 2024 will the average temperature recorded across all of the U.S. be below 65°F?

[Recording box, activated manually]

General Instructions

Thanks for recording your first voice message! This study will take approximately 30 minutes to complete. You will earn a reward of \$6.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to answer 15 questions on various topics. Questions will have two or three possible options. Exactly one of the options is the correct answer. For each question, you will be asked to record yourself once to give advice on the question and explain your reasoning.

We are interested in how you would give advice in an informal conversation:

- You should share an explanation behind your response.
- Your recording will be played to a few other participants who will have to respond to the same question.
- Other participants can win a bonus for selecting the correct answer.

Importantly:

- You should first read the question, think about your response, and then record your answer.
- The recording begins once you click “Start Recording”.

- After you click to submit a recording, it can take a little while to upload. We kindly ask you to be patient.

We ask you not to search the answers on the internet:

- We are interested in the explanations behind your answer.
- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.
- If you leave the browser tab of this survey, you will not be eligible for the \$6.00 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

Bonus Payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the 15 questions you have answered will be randomly chosen.
- You will receive the bonus of \$10 if the participant selected the correct answer.

One of the participants who listened to your answer will be randomly chosen. You should therefore give your explanation in a way that makes the other respondent most likely to select the correct answer!

Much like you, participants listening to your recordings will have a chance to win a bonus of \$10 if they select the right question in a randomly selected round. Moreover, participants listening to your recordings will be informed that you will receive a bonus if they select the correct answer.

This study will take approximately 30 minutes to complete. You will earn a reward of \$6.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will record a number of voice messages on different questions. Which one of the following statements is true?

- I should answer as if I'm talking to myself because the recording will never be played to another person.
- I should give a well-rehearsed response as if I'm giving a speech to a large audience of people that I don't know.
- I should share an explanation behind my response as if I were to give advice in an informal conversation. My response will be shared with other participants who later have to respond to the same question.

Which one of the following statements is true?

- On questions where I will be recorded, each recording only starts once I click "Start Recording" on a page.
- On questions where I will be recorded, each recording starts as soon as I enter a page.

Which one of the following statements is true?

- I can leave the tab of the experiment to search for answers online without consequences on my payment.
- If I leave the tab of the experiment, I will not be paid.

Remember!

Your chances of receiving the bonus payment are highest if the other participant chooses the correct answer.

Main Part: Example Task (Inflation)

On the next page, a question will be displayed. You should first read the question, think about your response, and then record your answer. The recording begins once you click “Start Recording”. After recording your advice, you will select your own answer to the question.

Read the question, then record your explanation!

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy:

1. More than today
2. Exactly the same as today
3. Less than today

Record an explanation that helps the other participant select the correct answer.

[Recording box, activated manually]

Provide your best answer

Please answer what you think is the correct answer to the question.

[Question with multiple choice response]

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Your answer to the following question will not affect your reward or bonus payment for this study, so please answer honestly. Did you search the answer to any of the 15 questions before providing your advice or your own answer?

- Yes

- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

2 Explanation receiver experiment

Welcome

In this survey, you will be asked questions on various topics. Besides giving your answer to a question, you will sometimes listen to a voice message of someone else's thoughts on the question. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.) and only works on Firefox and Chrome.

Voice Messages from Previous Participants

For some of the questions, you will first listen to a voice message from another participant. In a previous survey, we asked respondents to record their thoughts on the same questions that you will be asked.

Test Your Speaker

Use the play button below to test your speaker. Click "Play" to play back a voice message and select the sentence that you heard in the text box below.

[Player of a recording of someone saying "The dog runs in the park."]

Attention Check

Please select the sentence that you listened to in the voice message above:

- The koala climbs up the tree.
- The dog runs in the park.
- The lion looks at the gazelle.
- The cat waits for the mouse to come back.

- The fox sneaks through the garden.
- The turtle swims in the sea.

General Instructions

This study will take approximately 30 minutes to complete. You will earn a reward of \$6.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to answer 15 questions on various topics. Questions will have two or three possible options. Exactly one of the options is the correct answer. In each round, there are four steps:

1. You provide your best answer to the question.
2. You get information about a previous respondent's answer:
 - For some questions, you will listen to a voice message of another person once.
 - For other questions, you will see the answer of another participant to the question.
3. You have a second chance to provide your best answer to the question. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

When you enter a page with a recording, the recording will play automatically. You will only be able to listen to it once.

Bonus Payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the 15 rounds you have answered will be randomly chosen.

- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive the bonus of \$10 if you selected the correct answer.

Participants who made the recordings were informed they had a chance to win a bonus of \$10 if you selected the correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected the correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will listen to a number of voice messages on different questions. Which one of the following statements is true?

- Before each of the 15 questions I'll be asked, I will listen to a voice message from another respondent.
- Before each of the 15 questions I'll be asked, I will see what another respondent answered.
- On some of the 15 questions I'll be asked, I might listen to a voice message from another respondent. On other questions, I will see what the respondent answered.

Which one of the following statements is true?

- On questions with a voice recording, each recording only starts playing once I click "Play" on a page.
- On questions with a voice recording, each recording starts playing automatically when I enter a page, so I should read the question and pay attention to the recording.

Which one of the following statements is true?

- The answers I chose have no effect on my expected bonus.
- I maximize my expected bonus by selecting my best answer to each question.

Main Part: Example Question (Inflation)

[Prior Choice:] Provide Your Best Answer

Please answer what you think is the correct answer to the question.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy:

1. More than today
2. Exactly the same as today
3. Less than today

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

[Explanation Treatment:]

Now, you will listen to a recording of a voice message from a previous respondent who shares the explanation behind their answer to the exact same question that you just answered. The voice message will automatically start playing.

Please listen closely to the recording.

You will be able to proceed to the next page once the recording has finished playing.

[PAGEBREAK]

Listen to the other respondent's answer

[Box with question text]

Other person's answer:

[Answer of other respondent]

[Recording of other respondent, on auto-play]

[Choice Only Treatment:]

Now, you will observe the answer from a previous respondent.

Please pay close attention to the other person's answer.

[PAGEBREAK]

Read the other respondent's answer

[Box with question text]

Other person's answer:

[Answer of other respondent]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.

[Question with multiple choice answer]

Your answer is correct if you selected the right answer.

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Did you look up any answers on the internet? Your response to this question will not affect your payment. Please answer truthfully.

- Yes
- No

Additional Information

The explanations you just listened to likely differed systematically in how rich or sparse they were. Rich explanations include substantial details on the reasoning and tend to be elaborate, while sparse explanations provide limited details.

Which statement do you most agree with? Over the course of this experiment, I learned about whether a given answer is correct...

- ...more from sparse explanations than from rich explanations.
- ...more from rich explanations than from sparse explanations.
- ...equally much from rich and sparse explanations.

Why do you think this is the case?

[Open text box]

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

3 Transcript receiver experiment

Welcome

In this survey, you will be asked questions on various topics. This study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). If you are accessing this study on a smartphone, a tablet or any other non-PC devices, please switch to PC and enter the study again, or return the submission on Prolific.

Please write at least 15 words describing your opinion about daylight savings in the United States. Whether you are in favor or against daylight savings does not affect your eligibility to participate in this study. However, we ask that you write at least 15 words on your thoughts about this topic.

[Text box]

General Instructions

This study will take approximately 30 minutes to complete. You will earn a reward of \$6.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to answer 15 questions on various topics. Questions will have two or three possible options. Exactly one of the options is the correct answer. In each round, there are four steps:

1. You provide your best answer to the question.
2. You get information about a previous respondent's answer:
 - For some questions, you will see another participant's answer to the question and additionally read the transcript of the explanation they recorded.
 - For other questions, you will only see the participant's answer.
3. You have a second chance to provide your best answer to the question. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

When you enter a page with an explanation, the transcript will be displayed automatically. You will only be able to read it once

Bonus Payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the 15 rounds you have answered will be randomly chosen.
- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive the bonus of \$10 if you selected the correct answer.

Participants who made the recordings were informed they had a chance to win a bonus of \$10 if you selected the correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected the correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will read a number of explanations on different questions. Which one of the following statements is true?

- Before each of the 15 questions I'll be asked, I will read the transcript of an explanation recorded by a previous respondent.
- Before each of the 15 questions I'll be asked, I will see what a previous respondent answered.
- On some of the 15 questions I'll be asked, I might read the transcript of an explanation recorded by a previous respondent. On other questions, I will see what the previous respondent answered.

Which one of the following statements is true?

- Previous respondents typed their explanations into a text box.
- Previous respondents recorded their explanations as voice messages which were then automatically transcribed into text.

Which one of the following statements is true?

- The answers I chose have no effect on my expected bonus.
- I maximize my expected bonus by selecting my best answer to each question.

Main Part: Example Question (Inflation)

[Prior Choice:] Provide Your Best Answer

Please answer what you think is the correct answer to the question.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy:

1. More than today
2. Exactly the same as today
3. Less than today

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

[Transcript Treatment:]

Read the other respondent's explanation

[Box with question text]

Other person's answer:

[Answer of other respondent]

Other person's explanation:

[Transcript of explanation, displayed progressively]

[Choice Only Treatment:]

Read the other respondent's answer

[Box with question text]

Other person's answer:

[Answer of other respondent]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.

[Question with multiple choice answer]

Your answer is correct if you selected the right answer.

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Did you look up any answers on the internet? Your response to this question will not affect your payment. Please answer truthfully.

- Yes
- No

Additional Information

The explanations you just read likely differed systematically in how rich or sparse they were. Rich explanations include substantial details on the reasoning and tend to be elaborate, while sparse explanations provide limited details.

Which statement do you most agree with? Over the course of this experiment, I learned about whether a given answer is correct...

- ...more from sparse explanations than from rich explanations.
- ...more from rich explanations than from sparse explanations.
- ...equally much from rich and sparse explanations.

Why do you think this is the case?

[Open text box]

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

4 Choice & Confidence receiver experiment

Welcome

In this survey, you will be asked questions on various topics. This study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). If you are accessing this study on a smartphone, a tablet or any other non-PC devices, please switch to PC and enter the study again, or return the submission on Prolific.

Please write at least 15 words describing your opinion about daylight savings in the United States. Whether you are in favor or against daylight savings does not affect your eligibility to participate in this study. However, we ask that you write at least 15 words on your thoughts about this topic.

[Text box]

General Instructions

This study will take approximately 15 minutes to complete. You will earn a reward of \$3.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to answer 15 questions on various topics. Questions will have two or three possible options. Exactly one of the options is the correct answer. In each round, there are three steps:

1. You provide your best answer to the question.
2. You get information about a previous respondent's answer:

- For some questions, you will see another participant's answer to the question and the participant's confidence in their answer, on a scale from 0% to 100%. 0% means the respondent was not at all certain, 100% means the respondent was fully certain.
 - For other questions, you will only see the participant's answer.
3. You have a second chance to provide your best answer to the question. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

The confidence question previous respondents were asked was exactly like the confidence question you will be asked after giving your answers in each round.

Bonus Payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the 15 rounds you have answered will be randomly chosen.
- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive the bonus of \$10 if you selected the correct answer.

Participants who gave the previous answers were informed they had a chance to win a bonus of \$10 if you selected the correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected the correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will read a number of explanations on different questions. Which one of the following statements is true?

- Before each of the 15 questions I'll be asked, I will see what a previous respondent answered and how confident they were.
- Before each of the 15 questions I'll be asked, I will see what a previous respondent answered.
- On some of the 15 questions I'll be asked, I might see what a previous respondent answered and how confident they were. On other questions, I will see what the previous respondent answered.

Which one of the following statements is true?

- Confidence is expressed as a sentence such as "I am not very confident" or "I'm extremely confident."
- Confidence is expressed as a percentage from 0% to 100%.

Which one of the following statements is true?

- The answers I choose have no effect on my expected bonus.
- I maximize my expected bonus by selecting my best answer to each question.

Main Part: Example Question (Inflation)

[Prior Choice:] Provide Your Best Answer

Please answer what you think is the correct answer to the question.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy:

1. More than today
2. Exactly the same as today
3. Less than today

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

[Choice & Confidence Treatment:]

Read the other respondent's answer

[Box with question text]

Other person's answer:

[Answer of other respondent]

Other person's confidence:

[Confidence of other respondent, from 0% to 100%]

[Choice Only Treatment:]

Read the other respondent's answer

[Box with question text]

Other person's answer:

[Answer of other respondent]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.

[Question with multiple choice answer]

Your answer is correct if you selected the right answer.

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Did you look up any answers on the internet? Your response to this question will not affect your payment. Please answer truthfully.

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

5 Richness manipulation receiver experiment

Welcome

In this survey, you will be asked questions on various topics. This study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). If you are accessing this study on a smartphone, a tablet or any other non-PC devices, please switch to PC and enter the study again, or return the submission on Prolific.

Please write at least 15 words describing your opinion about daylight savings in the United States. Whether you are in favor or against daylight savings does not affect your eligibility to participate in this study. However, we ask that you write at least 15 words on your thoughts about this topic.

[Text box]

Please also complete the captcha below.

[Captcha]

Attention Check

This question is an attention check. To be eligible to participate in this study, we simply ask you to record a voice message that lasts at least 20 seconds on your opinion about daylight savings in the United States. Whether you are in favor or against daylight savings does not affect your eligibility to participate in this study. However, we ask that you record a message that lasts at least 20 seconds.

[Recording box, activated manually]

General Instructions

This study will take approximately 25 minutes to complete. You will earn a reward of \$4.50 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to answer 15 questions on various topics. Questions will have two or three possible options. Exactly one of the options is the correct answer. In each round, there are three steps:

1. You provide your best answer to the question.
2. You will see another participant's answer to the question and additionally read an edited version of the explanation they recorded.
3. You have a second chance to provide your best answer to the question. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

When you enter a page with an explanation, the explanation will be displayed automatically. You will only be able to read it once.

We ask you not to search the answers on the internet or to use external tools:

- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.

- If you use the internet or external tools, you will not be eligible for the \$4.50 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

Bonus Payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the 15 rounds you have answered will be randomly chosen.
- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive the bonus of \$10 if you selected the correct answer.

Participants who made the recordings were informed they had a chance to win a bonus of \$10 if you selected the correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected the correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will read a number of explanations on different questions. Which one of the following statements is true?

- Before each of the 15 questions I'll be asked, I will read an explanation written by the researchers behind this survey.
- Before each of the 15 questions I'll be asked, I will only see what a previous respondent answered.
- Before each of the 15 questions I'll be asked, I will read an edited version of an explanation recorded by a previous respondent.

Which one of the following statements is true?

- Previous respondents typed their explanations into a text box.
- Previous respondents recorded their explanations as voice messages which were then automatically transcribed and edited.

Which one of the following statements is true?

- The answers I choose have no effect on my expected bonus.
- I maximize my expected bonus by selecting my best answer to each question.

Main Part: Example Question (Inflation)

[Prior Choice:] Provide Your Best Answer

Please answer what you think is the correct answer to the question.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy:

1. More than today
2. Exactly the same as today
3. Less than today

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

[Rich or Sparse Version Treatment:]

Read the other respondent's explanation

[Box with question text]

Other person's answer:

[Answer of other respondent]

Other person's explanation:

[Transcript of explanation, displayed progressively]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.

[Question with multiple choice answer]

Your answer is correct if you selected the right answer.

How certain are you that your above answer is correct?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Did you look up any answers on the internet? Your response to this question will not affect your payment. Please answer truthfully.

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

6 Continuous balls-and-urns orator experiment

Welcome

In this survey, we will ask you to record voice messages to answer different questions. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). Please make sure you are in a quiet environment. You will only receive your completion payment if your voice is clearly recorded.

You are eligible for a bonus of \$10 for one of your recordings! The details of how you might receive a bonus will be explained on the following pages.

For each question, your recording will actually be played to another respondent. The other respondent will have to answer the same question as you after listening to your recording.

Privacy & Anonymity

All voice messages are treated strictly anonymously. They will never be linked to your person and will never be published anywhere. This data will be used solely for academic research. You can therefore talk freely and informally in each voice message. The other respondent who will listen to your voice recording will not be given any more information about you: your anonymity will be preserved.

Test Your Microphone

Use the recorder below to test your microphone. Click “Record”, say the sentence “The dog runs in the park.”, then click “Stop Recording & Submit”. You may have to give your browser permission to access the microphone after you click “Record”. After a recording, it might take the website a few seconds to upload your recording: please be patient.

Trial Question

This trial question is an attention check. To be eligible to participate in this study, we simply ask you to record a voice message that lasts at least 20 seconds on your thoughts about this topic. There is no correct answer, and it does not matter what you say, as long as you record a message that lasts at least 20 seconds.

On how many days in 2025 will the average temperature recorded across all of the U.S. be below 65°F?

[Recording box, activated manually]

General Instructions

Thanks for recording your first voice message! This study will take approximately 10 minutes to complete. You will earn a reward of \$2.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

We are interested in how you would give advice in an informal conversation:

- You should share an explanation behind your response.
- Your recording will be played to a few other participants who will have to respond to the same question.
- Other participants can win a bonus for selecting the correct answer.

Importantly:

- You should first read the question, think about your response, and then record your answer.
- The recording begins once you click “Start Recording”.
- After you click to submit a recording, it can take a little while to upload. We kindly ask you to be patient.

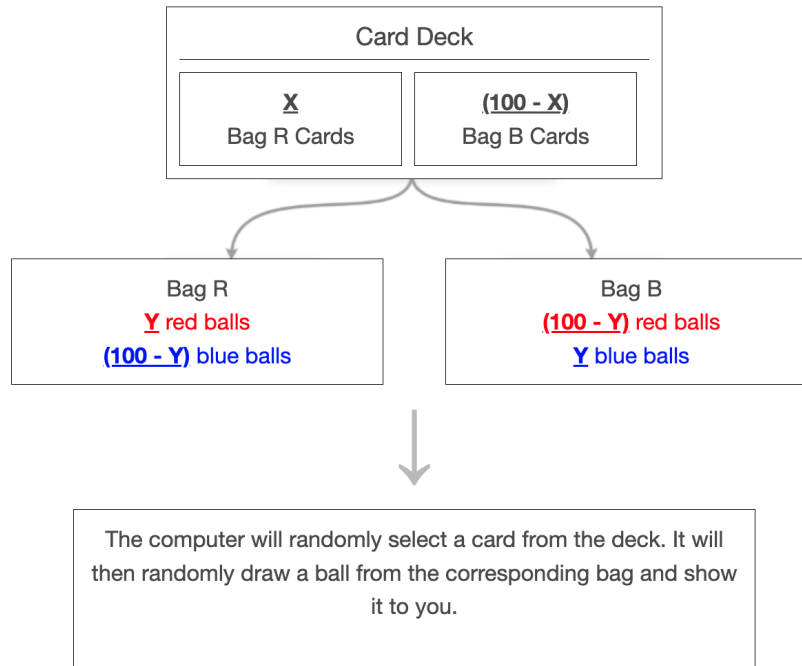
We ask you not to search the answers on the internet:

- We are interested in the explanations behind your answer.
- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.
- If you leave the browser tab of this survey, you will not be eligible for the \$2.00 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

The Task

In this study, your task is to guess which of two bags was secretly selected by a computer. As a hint, you will see a colored ball that was drawn from the secretly-selected bag.

The setup is as follows:



- There is a deck of 100 cards, some of which have “Bag R” written on them and some of which have “Bag B” written on them. We denote the number of cards with Bag R on them by “ X ” (the number of cards with Bag B is $100 - X$).
- There are two bags, Bag R and Bag B. Both bags contain 100 balls each, some of which are red and some of which are blue. Bag R always contains at least as many red balls as blue balls and Bag B always contains at least as many blue balls as red balls. It is always the case that the number of red balls in Bag R is equal to the number of blue balls in Bag B. We denote this number by “ Y ”.
- For example, Bag R might contain 95 red balls and 5 blue balls, and so Bag B would contain 5 red balls and 95 blue balls.

The task proceeds as follows:

1. You will be told how many cards have "Bag R" vs. "Bag B" written on them.
2. You will be told how many balls in Bag R and Bag B are red or blue.
3. The computer will randomly select one of the 100 cards. If the card has "Bag R" written on it, the computer selects Bag R. If the card has "Bag B" written on it, the computer selects Bag B. You will not observe which card was drawn, so you will not know for sure which bag was selected.
4. The computer will then randomly draw one ball from the selected bag and show it to you.
5. You will then be asked to provide a percentage chance to indicate how likely you think it is that the computer selected Bag R or B.

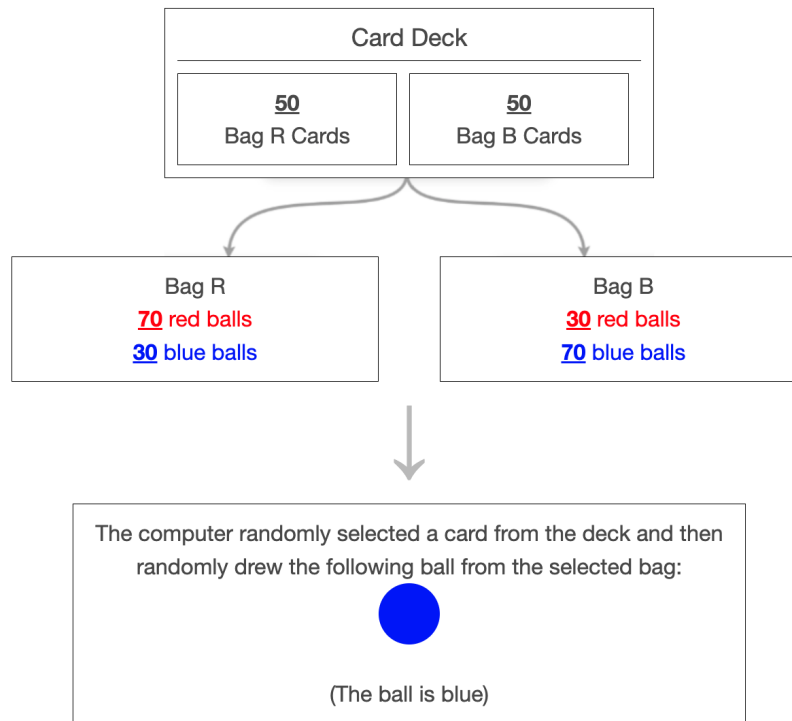
Example

- In the example above, there are 50 cards labeled "Bag R" and 50 cards labeled "Bag B". Bag R contains 70 red and 30 blue balls and bag B contains 30 red and 70 blue balls.
- A blue ball was drawn.
- You would then tell us the percentage chance you think the computer selected Bag R, based on the information provided.

Your certainty

During the task:

- You will tell us the percentage chance you think the computer selected Bag R.
- We will ask you how certain you are about your answer. Specifically, we are interested in how likely you think it is (in percentage terms) that your answer is within ± 3 percentage points of the statistically correct answer.



Bonus payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the participants who listened to your answer will be randomly chosen.
- You will receive \$10 if the participant's answer is within ± 3 percentage points of the statistically correct answer.
- You should therefore give your explanation in a way that makes the other respondent most likely to select the correct answer!

Much like you, participants listening to your recordings will have a chance to win a bonus of \$10 if their answer is within ± 3 percentage points of the statistically correct answer. Moreover, participants listening to your recordings will be informed that you will receive a bonus if they select an answer within ± 3 percentage points of the statistically correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will record a voice message on a question. Which one of the following statements is true?

- I should answer as if I'm talking to myself because the recording will never be played to another person.
- I should give a well-rehearsed response as I'm giving a speech to a large audience of people that I don't know.
- I should share an explanation behind my response as if I were to give advice in an informal conversation. My response will be shared with other participants who later have to respond to the same question.

Which one of the following statements is true?

- On questions where I will be recorded, each recording only starts once I click "Start Recording" on a page.
- On questions where I will be recorded, each recording starts as soon as I enter a page.

Which one of the following statements is true?

- I can leave the tab of the experiment to search for answers online without consequences on my payment.
- If I leave the tab of the experiment, I will not be paid.

Which one of the following statements is true?

- When I'm asked to indicate my certainty about my decision, the people running this study are interested in how certain I am that my estimate of the percentage chance that Bag R was selected is statistically correct.

- When I'm asked to indicate my certainty about my decision, the people running this study are interested in how certain I am that Bag R was selected.

Which one of the following statements is true?

- If the computer draws a card with "R" on it from the deck of cards, it will draw a ball from Bag R and show it to me.
- If the computer draws a card with "R" on it from the deck of cards, it will draw a ball from a randomly selected bag and show it to me.

Which one of the following statements is NOT true?

- Bags R and B always have the same proportion of red and blue balls.
- Bag R always has at least as many red balls as blue balls.
- Bag B always has at least as many blue balls as red balls.

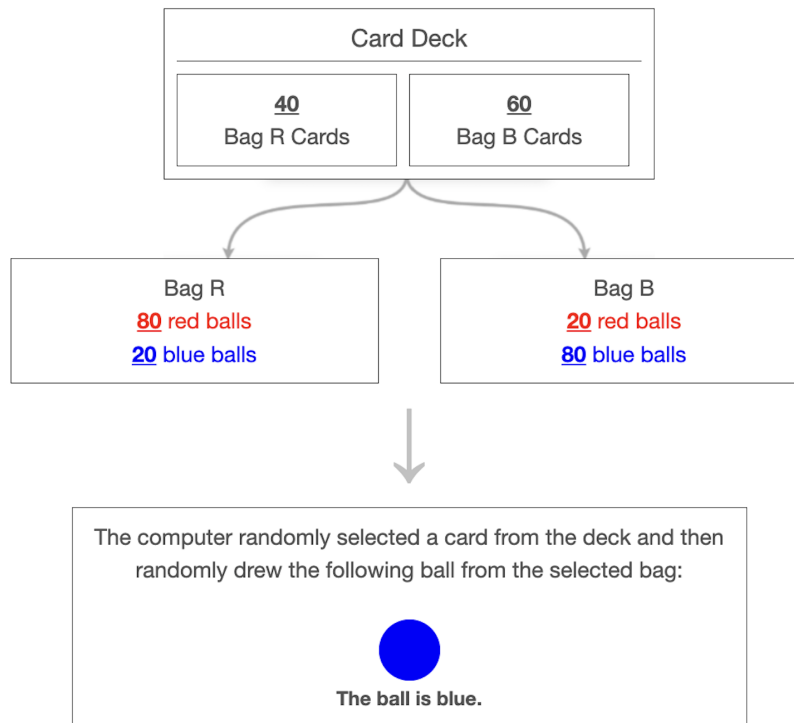
Remember!

Your chances of receiving the bonus payment are highest if the other participant chooses the correct answer.

Main Part

On the next page, the main task will be displayed. You should first read the question, think about your response and then record your answer. The recording begins once you click "Start Recording". After recording your explanation, you will select your own answer to the question.

Read the question, then record your explanation!



Given that this ball was drawn, how likely do you think it is that Bag R (as opposed to Bag B) has been selected?

Record an explanation that helps the other participant select the correct answer.

[Recording box, activated manually]

Provide your best answer

Please answer what you think is the correct answer to the question.

Given that this ball was drawn, how likely do you think it is that Bag R (as opposed to Bag B) has been selected?

[Input X]% likely that Bag R was selected.

[$100 - X$]% likely that Bag B was selected.

How certain are you that your above answer is within ± 3 percentage points of the statistically correct answer?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Your answer to the following question will not affect your reward or bonus payment for this study, so please answer honestly. Did you search the answer to the question or use external tools, like an AI assistant, before providing your advice or your own answer?

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

7 Continuous balls-and-urns receiver experiment

Welcome

In this survey, you will be asked to make an estimate. Besides providing your estimate, you might listen to a voice message of someone else's thoughts on the correct estimate. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.) and only works on Firefox and Chrome.

Voice messages from previous participants

Before providing your estimate, you may be asked to first listen to a voice message from another participant. In a previous survey, we asked respondents to record their thoughts on the same estimation task.

Test your speaker

Use the play button below to test your speaker. Click "Play" to play back a voice message and select the sentence that you heard in the text box below.

[Play button screenshot here]

Please select the sentence that you listened to in the voice message above:

- The koala climbs up the tree.

- The dog runs in the park.
- The lion looks at the gazelle.
- The cat waits for the mouse to come back.
- The fox sneaks through the garden.
- The turtle swims in the sea.

General Instructions

This study will take approximately 10 minutes to complete. You will earn a reward of \$2 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to complete an estimation task. In each round, there are four steps:

1. You provide your best answer to the task.
2. You get information about a previous respondent's answer:
 - You might listen to a voice message of another participant once.
 - You might see the answer of another participant.
3. You have a second chance to provide your best answer to the task. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

When you enter a page with a recording, the recording will play automatically. You will only be able to listen to it once.

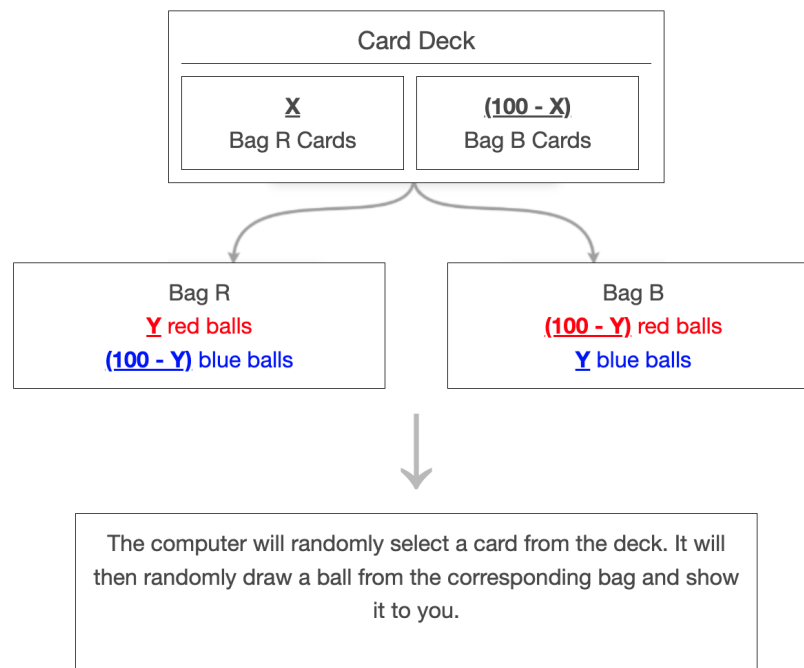
We ask you not to search the answers on the internet or to use external tools:

- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.

- If you leave the browser tab of this survey, you will not be eligible for the \$2 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

The Task

In this study, your task is to guess which of two bags was secretly selected by a computer. As a hint, you will see a colored ball that was drawn from the secretly-selected bag.



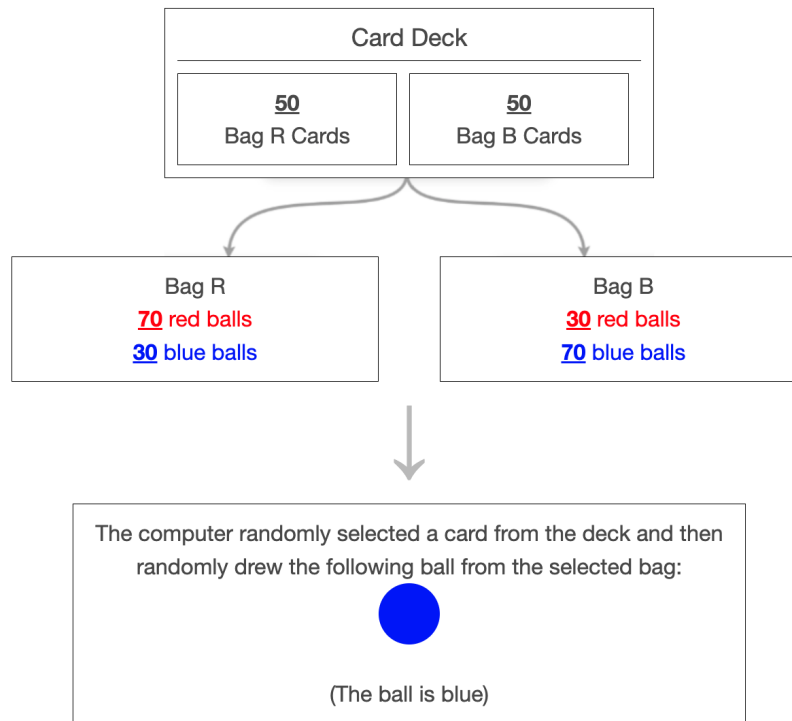
- There is a deck of 100 cards, some of which have “Bag R” written on them and some of which have “Bag B” written on them. We denote the number of cards with Bag R on them by “ X ” (the number of cards with Bag B on them is $100 - X$).
- There are two bags, Bag R and Bag B. Both bags contain 100 balls each, some of which are red and some of which are blue. Bag R always contains at least as many red balls as blue balls, and Bag B always contains at least as many blue balls as red balls. It is always the case that the number of red balls in Bag R is equal to the number of blue balls in Bag B. We denote this number by “ Y ”.

- For example, Bag R might contain 95 red balls and 5 blue balls, and so Bag B would contain 5 red balls and 95 blue balls.

The task proceeds as follows:

1. You will be told how many cards have "Bag R" vs. "Bag B" written on them.
2. You will be told how many balls in Bag R and Bag B are red or blue.
3. The computer will randomly select one of the 100 cards. If the card has "Bag R" written on it, the computer selects Bag R. If the card has "Bag B" written on it, the computer selects Bag B. You will not observe which card was drawn, so you will not know for sure which bag was selected.
4. The computer will then randomly draw one ball from the selected bag and show it to you.
5. You will then be asked to provide a percentage chance to indicate how likely you think it is that the computer selected Bag R or B.

Example



- In the example above, there are 50 cards labeled "Bag R" and 50 cards labeled "Bag B". Bag R contains 70 red and 30 blue balls and Bag B contains 30 red and 70 blue balls.
- A blue ball was drawn.
- Step 1: You enter the percentage chance you think the computer selected Bag R, based on the information provided.
- Step 2: You see a randomly drawn other participant's answer, and you may listen to their explanation. Note: that participant saw exactly the same problem, including the same draw of the ball.
- Step 3: You enter the percentage chance you think the computer selected Bag R, based on the information provided and the other participant's answer.

Your certainty

During the task:

- You will tell us the percentage chance you think the computer selected Bag R.
- We will ask you how certain you are about your answer. Specifically, we are interested in how likely you think it is (in percentage terms) that your answer is within ± 3 percentage points of the statistically correct answer.

Bonus payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive \$10 if your answer is within ± 3 percentage points of the statistically correct answer.

Participants who made the recordings were informed they had a chance to win a bonus of \$10 if you selected an answer within ± 3 percentage points of the statistically correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected an answer within ± 3 percentage points of the statistically correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

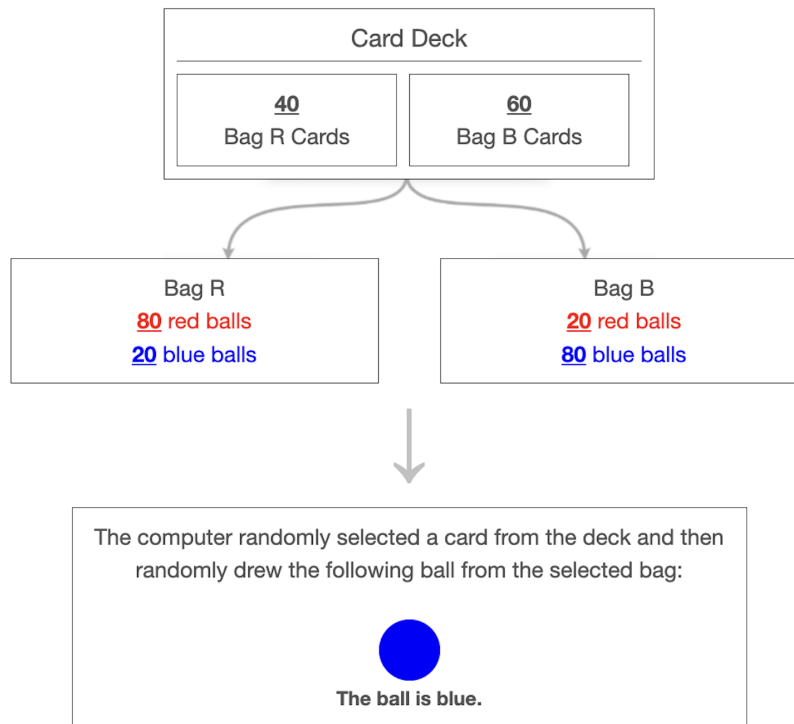
1. In this study, you will listen to a number of voice messages on different questions. Which one of the following statements is true?
 - I will listen to a voice message from another respondent about the task.
 - I will see what another respondent answered on the task.

- I might listen to a voice message from another respondent about the task, or see what another respondent answered on the task.
2. Which one of the following statements is true?
- On a page with a voice recording, the recording only starts playing once I click “Play”.
 - On a page with a voice recording, the recording starts playing automatically when I enter, so I should read the question and pay attention to the recording.
3. Which one of the following statements is true?
- The answers I choose have no effect on my expected bonus.
 - I maximize my expected bonus by selecting my best answer to each question.
4. Which one of the following statements is true?
- When I’m asked to indicate my certainty about my decision, the people running this study are interested in how certain I am that my estimate of the percentage chance that Bag R was selected is statistically correct.
 - When I’m asked to indicate my certainty about my decision, the people running this study are interested in how certain I am that Bag R was selected.
5. Which one of the following statements is true?
- If the computer draws a card with “R” on it from the deck of cards, it will draw a ball from Bag R and show it to me.
 - If the computer draws a card with “R” on it from the deck of cards, it will draw a ball from a randomly selected bag and show it to me.
6. Which one of the following statements is NOT true?
- Bags R and B always have the same proportion of red and blue balls.
 - Bag R always has at least as many red balls as blue balls.
 - Bag B always has at least as many blue balls as red balls.

Main Part

On the next page, the main task will be displayed.

[Prior Choice:] Provide your best answer



Given that this ball was drawn, how likely do you think it is that Bag R (as opposed to Bag B) has been selected?

[Input X]% likely that Bag R was selected.

[100 – X]% likely that Bag B was selected.

How certain are you that your above answer is within +/- 3 percentage points of the statistically correct answer?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

[Explanation Treatment:]

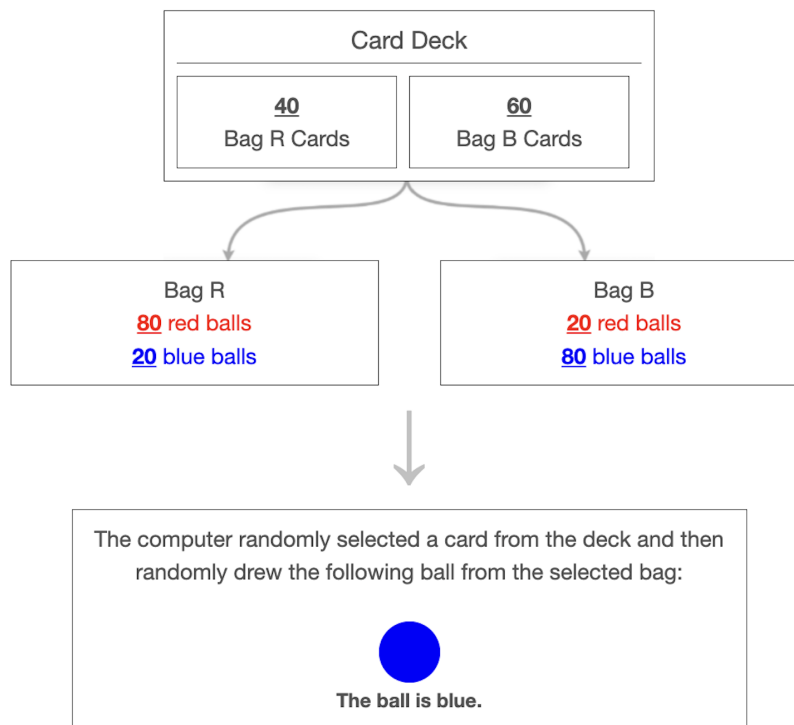
Now, you will listen to a recording of a voice message from a previous respondent who shares the explanation behind their answer to the exact same question that you just answered. The voice message will automatically start playing.

Please listen closely to the recording.

You will be able to proceed to the next page once the recording has finished playing.

[PAGEBREAK]

Listen to the other respondent's answer



Other person's answer:

[Answer of other respondent]

[Recording of other respondent, on auto-play]

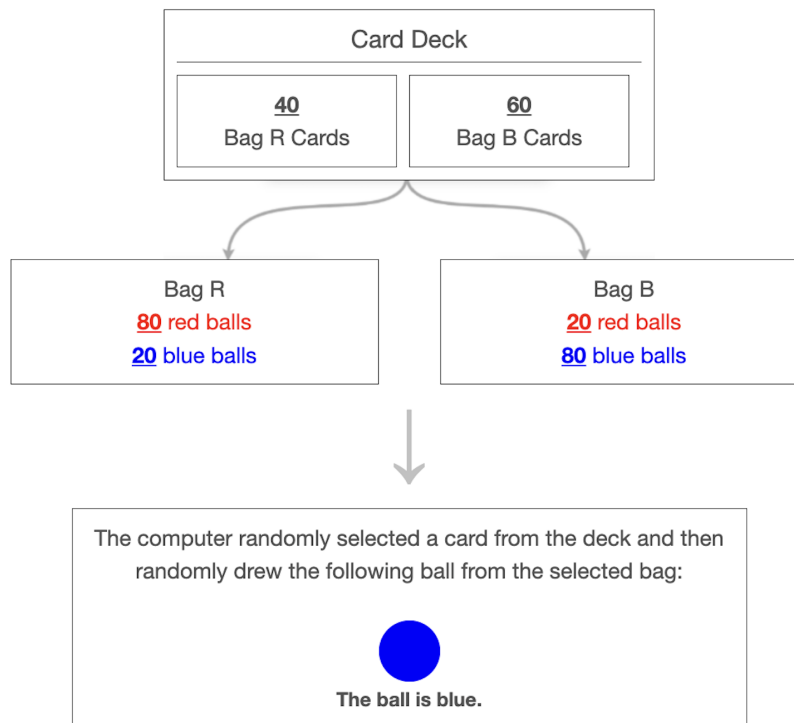
[Choice Only Treatment:]

Now, you will observe the answer from a previous respondent.

Please pay close attention to the other person's answer.

[PAGEBREAK]

Read the other respondent's answer



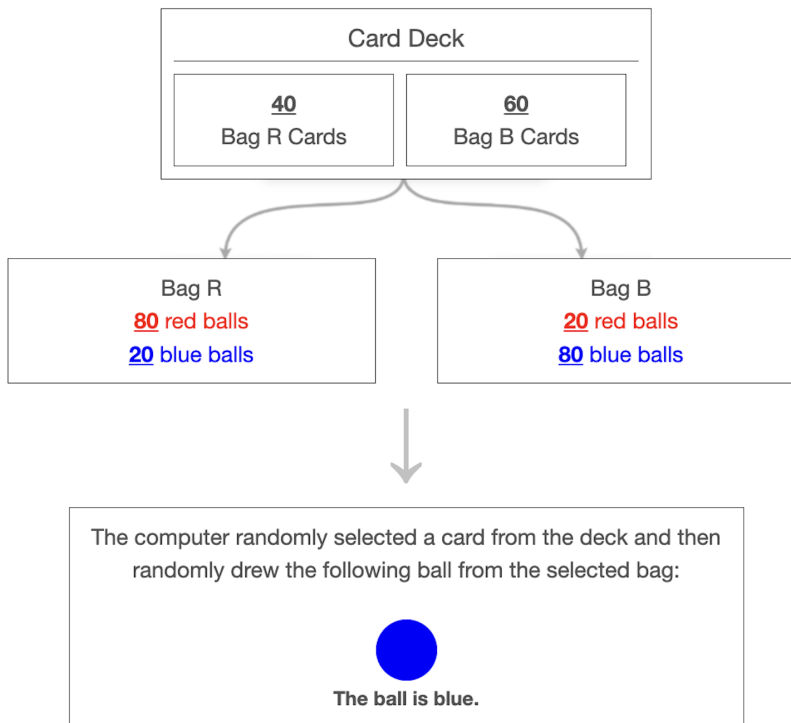
Other person's answer:

[Answer of other respondent]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.



Given that this ball was drawn, how likely do you think it is that Bag R (as opposed to Bag B) has been selected?

[Input X]% likely that Bag R was selected.

[100 – X]% likely that Bag B was selected.

How certain are you that your above answer is within +/- 3 percentage points of the statistically correct answer?

[Slider from 0% (Not at all certain) to 100% (Fully certain)]

Additional Questions

Your answer to the following question will not affect your reward or bonus payment for this study, so please answer honestly.

Did you search the answer to the question or use external tools, like an AI assistant, before providing your advice or your own answer?

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

8 Binary Balls-and-urns orator experiment

Welcome

In this survey, we will ask you to record voice messages to answer different questions. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). Please make sure you are in a quiet environment. You will only receive your completion payment if your voice is clearly recorded.

You are eligible for a bonus of \$10 for one of your recordings! The details of how you might receive a bonus will be explained on the following pages.

For each question, your recording will actually be played to another respondent. The other respondent will have to answer the same question as you after listening to your recording.

Privacy & Anonymity

All voice messages are treated strictly anonymously. They will never be linked to your person and will never be published anywhere. This data will be used solely for academic research. You can therefore talk freely and informally in each voice message. The other respondent who will listen to your voice recording will not be given any more information about you: your anonymity will be preserved.

Test Your Microphone

Use the recorder below to test your microphone. First click “Grant microphone access” and allow Qualtrics to access your microphone. Click “Record”, say the sentence “The

dog runs in the park.”, then click “Stop Recording & Submit”. You may have to give your browser permission to access the microphone after you click “Record”. After a recording, it might take the website a few seconds to upload your recording: please be patient.

Trial Question

This trial question is an attention check. To be eligible to participate in this study, we simply ask you to record a voice message that lasts at least 20 seconds on your thoughts about this topic. There is no correct answer, and it does not matter what you say, as long as you record a message that lasts at least 20 seconds.

On how many days in 2026 will the average temperature recorded across all of the U.S. be below 65°F?

[Recording box, activated manually]

General Instructions

Thanks for recording your first voice message! This study will take approximately 10 minutes to complete. You will earn a reward of \$2.00 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to complete a task. You will be asked to record yourself once to give advice on the task and explain your reasoning.

We are interested in how you would give advice in an informal conversation:

- You should share an explanation behind your response.
- Your recording will be played to a few other participants who will have to respond to the same question.
- Other participants can win a bonus for selecting a correct answer.

Importantly:

- You should first read the question, think about your response and then record your answer.
- The recording begins once you click “Start Recording”.
- After you click to submit a recording, it can take a little while to upload. We kindly ask you to be patient.

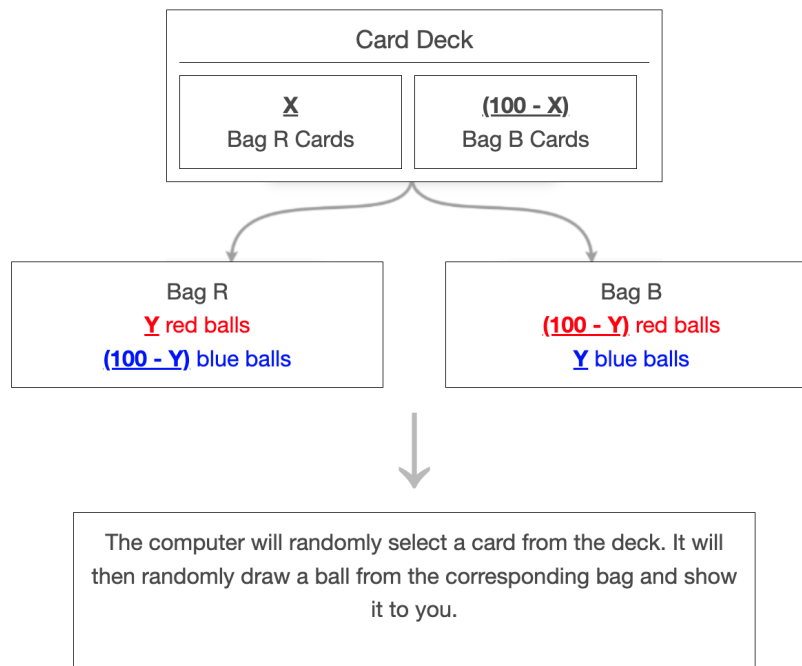
We ask you not to search the answers on the internet or to use external tools:

- We are interested in the explanations behind your answer.
- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.
- If you leave the browser tab of this survey, you will not be eligible for the \$2.00 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

The Task

In this study, your task is to guess which of two bags was secretly selected by a computer. As a hint, you will see a colored ball that was drawn from the secretly-selected bag.

The setup is as follows:



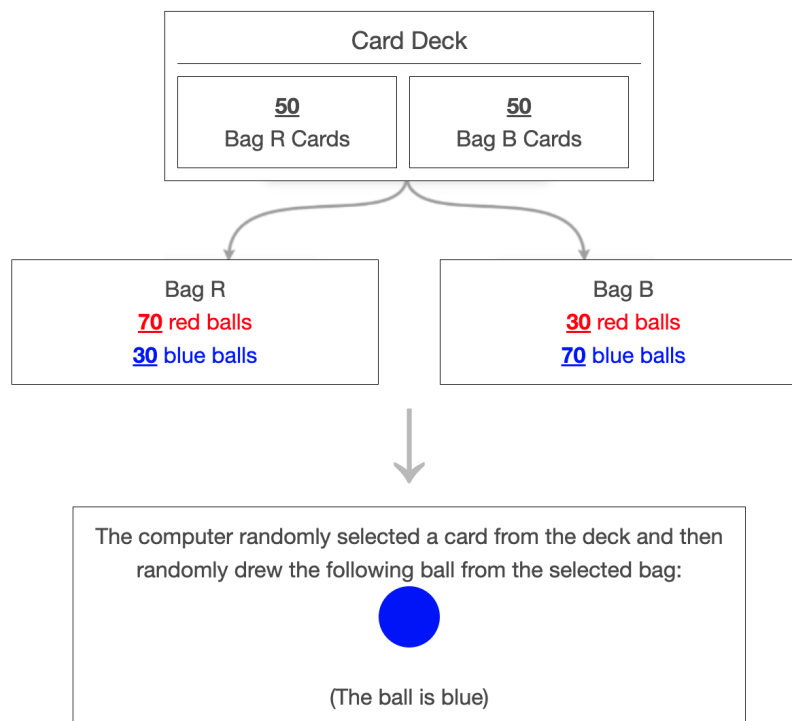
- There is a deck of 100 cards, some of which have “Bag R” written on them and some of which have “Bag B” written on them. We denote the number of cards with Bag R on them by “ X ” (the number of cards with Bag B on them is $100 - X$).
- There are two bags, Bag R and Bag B. Both bags contain 100 balls each, some of which are red and some of which are blue. Bag R always contains at least as many red balls as blue balls, and Bag B always contains at least as many blue balls as red balls. It is always the case that the number of red balls in Bag R is equal to the number of blue balls in Bag B. We denote this number by “ Y ”.
- For example, Bag R might contain 95 red balls and 5 blue balls, and so Bag B would contain 5 red balls and 95 blue balls.

The task proceeds as follows:

1. You will be told how many cards have “Bag R” vs. “Bag B” written on them.
2. You will be told how many balls in Bag R and Bag B are red or blue.

3. The computer will randomly select one of the 100 cards. If the card has “Bag R” written on it, the computer selects Bag R. If the card has “Bag B” written on it, the computer selects Bag B. You will not observe which card was drawn, so you will not know for sure which bag was selected.
4. The computer will then randomly draw one ball from the selected bag and show it to you.
5. You will then be asked to state whether you think Bag R or Bag B is more likely to have been selected.

Example



- In the example above, there are 50 cards labeled “Bag R” and 50 cards labeled “Bag B”. Bag R contains 70 red and 30 blue balls and Bag B contains 30 red and 70 blue balls.

- A blue ball was drawn.
- You would then tell us whether you think Bag R or Bag B is more likely to have been selected, based on the information provided.

Your certainty

During the task:

- You will tell us whether you think Bag R or Bag B is more likely to have been selected.
- We will ask you how certain you are about your answer. Specifically, we are interested in how likely you think it is (in percentage terms) that your answer is the statistically correct answer.

Bonus payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- One of the participants who listened to your answer will be randomly chosen.
- You will receive \$10 if the participant's answer is statistically correct.
- You should therefore give your explanation in a way that makes the other respondent most likely to select the correct answer!

Much like you, participants listening to your recordings will have a chance to win a bonus of \$10 if their answer is statistically correct. Moreover, participants listening to your recordings will be informed that you will receive a bonus if they select the statistically correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will record a voice message on a question. Which one of the following statements is true?

- I should answer as if I'm talking to myself because the recording will never be played to another person.
- I should give a well-rehearsed response as I'm giving a speech to a large audience of people that I don't know.
- I should share an explanation behind my response as if I were to give advice in an informal conversation. My response will be shared with other participants who later have to respond to the same question.

Which one of the following statements is true?

- On questions where I will be recorded, each recording only starts once I click "Start Recording" on a page.
- On questions where I will be recorded, each recording starts as soon as I enter a page.

Which one of the following statements is true?

- I can leave the tab of the experiment to search for answers online without consequences on my payment.
- If I leave the tab of the experiment, I will not be paid.

Which one of the following statements is true?

- I will be asked whether Bag R or Bag B is more likely to have been selected.
- I will be asked whether the ball is more likely to be blue or red.

Which one of the following statements is true?

- If the computer draws a card with "R" on it from the deck of cards, it will draw a ball from Bag R and show it to me.
- If the computer draws a card with "R" on it from the deck of cards, it will draw a ball from a randomly selected bag and show it to me.

Which one of the following statements is NOT true?

- Bags R and B always have the same proportion of red and blue balls.
- Bag R always has at least as many red balls as blue balls.
- Bag B always has at least as many blue balls as red balls.

Remember!

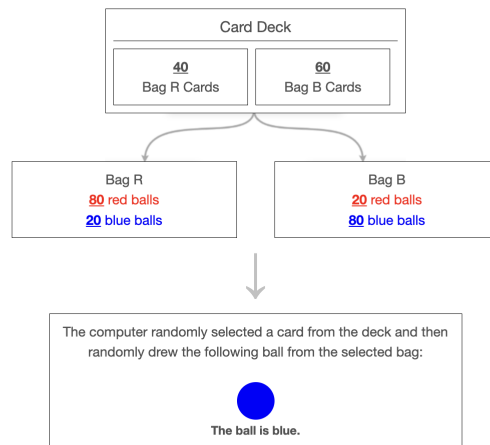
Your chances of receiving the bonus payment are highest if the other participant chooses the correct answer.

Main Part

On the next page, the main task will be displayed. You should first read the question, think about your response and then record your answer. The recording begins once you click “Start Recording”. After recording your explanation, you will select your own answer to the question.

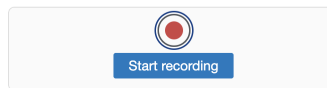
Read the question, then record your explanation!

Read the question, then record your explanation!



Given that this ball was drawn, **how likely** do you think it is that Bag R (as opposed to Bag B) has been selected?

Record an explanation that helps the other participant select the correct answer.



Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

Record an explanation that helps the other participant select the correct answer.

[Recording box, activated manually]

Provide your best answer

Please answer what you think is the correct answer to the question.

Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

- Bag R
- Bag B

How certain are you that your above answer is statistically correct?

[Slider from 0% (Very uncertain) to 100% (Completely certain)]

Additional Questions

Your answer to the following question will not affect your reward or bonus payment for this study, so please answer honestly. Did you search the answer to the question or use external tools, like an AI assistant, before providing your advice or your own answer?

- Yes
- No

[Followed, on separate pages, by elicitation of sex, age, ethnicity, accent, education, employment and political affiliation.]

9 Binary Balls-and-urns receiver experiment

Welcome

In this survey, you will be asked to make a decision. The study is designed for computer (PC or Mac) users only (desktop, laptop, etc.) and only works on Firefox and Chrome.

Voice messages from previous participants

Besides providing your decision, you might listen to a voice message of someone else's thoughts on the correct answer. In a previous survey, we asked respondents to record their thoughts on the same decision task.

Test your speaker

Use the play button below to test your speaker. Click "Play" to play back a voice message and select the sentence that you heard below.

[Play button screenshot here]

Please select the sentence that you listened to in the voice message above:

- The koala climbs up the tree.
- The dog runs in the park.

- The lion looks at the gazelle.
- The cat waits for the mouse to come back.
- The fox sneaks through the garden.
- The turtle swims in the sea.

General Instructions

This study will take approximately 10 minutes to complete. You will earn a reward of \$2 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Survey Structure

In this study, you will be asked to complete a task. There are three steps:

1. You provide your best answer to the task.
2. You get information about a previous respondent's answer:
 - You might listen to a voice message of another participant once.
 - You might see the answer of another participant.
3. You have a second chance to provide your best answer to the task. Your answer may or may not be different from your response in (1), given what you learned about the other participant's answer in (2).

When you enter a page with a recording, the recording will play automatically. You will only be able to listen to it once.

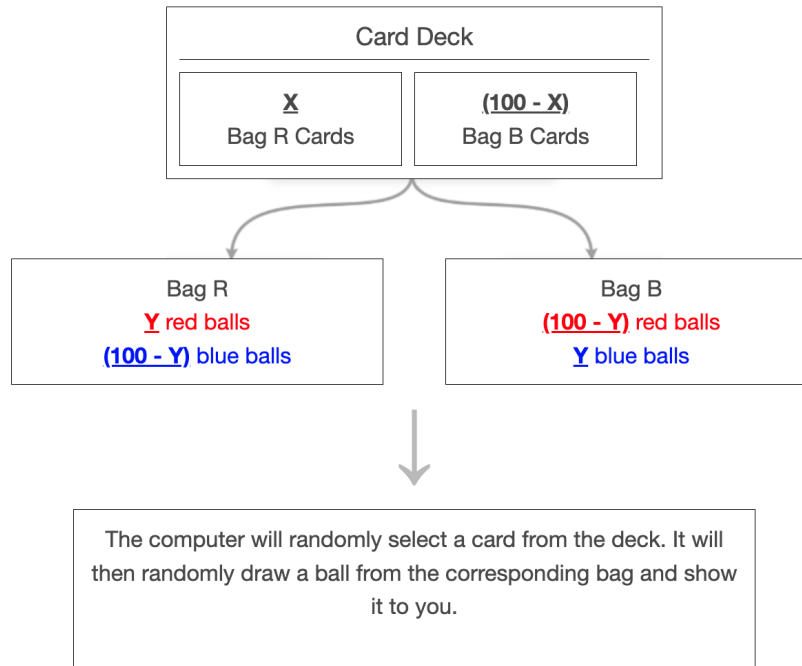
We ask you not to search the answers on the internet or to use external tools:

- To confirm that you do not search for answers, the survey will monitor whether the survey window remains active.
- If you leave the browser tab of this survey, you will not be eligible for the \$2 reward.
- You should remain focused on the survey window and answer questions as best you can using your previous knowledge.

The Task

In this study, your task is to guess which of two bags was secretly selected by a computer. As a hint, you will see a colored ball that was drawn from the secretly-selected bag.

The setup is as follows:

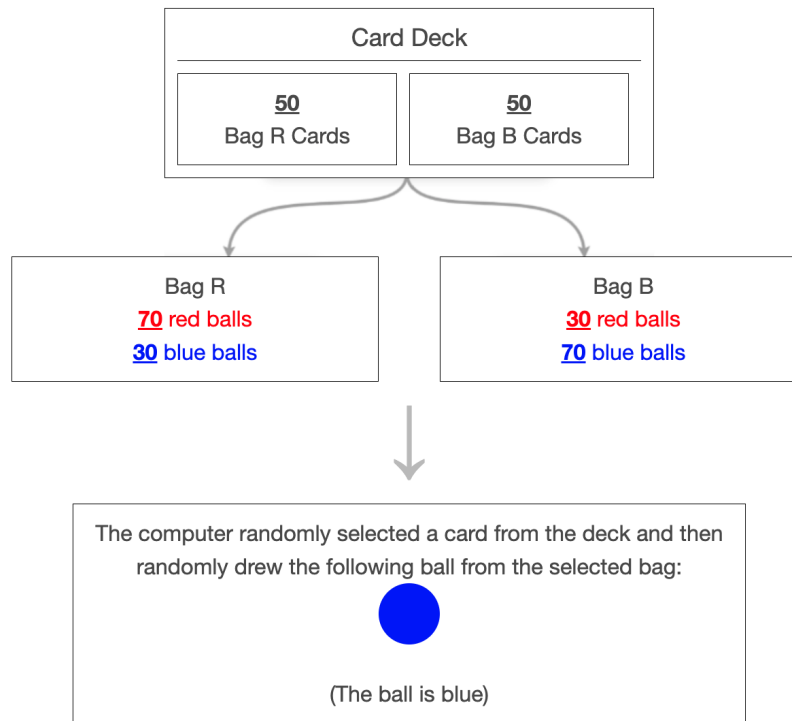


- There is a deck of 100 cards, some of which have “Bag R” written on them and some of which have “Bag B” written on them. We denote the number of cards with Bag R on them by “ X ” (the number of cards with Bag B on them is $100 - X$).
- There are two bags, Bag R and Bag B. Both bags contain 100 balls each, some of which are red and some of which are blue. Bag R always contains at least as many red balls as blue balls, and Bag B always contains at least as many blue balls as red balls. It is always the case that the number of red balls in Bag R is equal to the number of blue balls in Bag B. We denote this number by “ Y ”.
- For example, Bag R might contain 95 red balls and 5 blue balls, and so Bag B would contain 5 red balls and 95 blue balls.

The task proceeds as follows:

1. You will be told how many cards have “Bag R” vs. “Bag B” written on them.
2. You will be told how many balls in Bag R and Bag B are red or blue.
3. The computer will randomly select one of the 100 cards. If the card has “Bag R” written on it, the computer selects Bag R. If the card has “Bag B” written on it, the computer selects Bag B. You will not observe which card was drawn, so you will not know for sure which bag was selected.
4. The computer will then randomly draw one ball from the selected bag and show it to you.
5. You will then be asked to state whether you think Bag R or Bag B is more likely to have been selected.

Example



- In the example above, there are 50 cards labeled “Bag R” and 50 cards labeled “Bag B”. Bag R contains 70 red and 30 blue balls and Bag B contains 30 red and 70 blue balls.
- A blue ball was drawn.
- Step 1: You state whether you think Bag R or Bag B is more likely to have been selected, based on the information provided.
- Step 2: You see a randomly drawn other participant’s answer, and you may listen to their explanation. Note: that participant saw exactly the same problem, including the same draw of the ball.
- Step 3: You again state whether you think Bag R or Bag B is more likely to have been selected, based on the information provided and the other participant’s answer.

Your certainty

During the task:

- You will tell us whether you think Bag R or Bag B is more likely to have been selected.
- We will ask you how certain you are about your answer. Specifically, we are interested in how likely you think it is, in percentage terms, that your answer is the statistically correct answer.

Bonus payment

At the end of the survey, one out of every ten participants is randomly selected to be eligible for an additional bonus of up to \$10. If you are selected for the bonus payment:

- Either your answer from step (1) or your answer from step (3) will be randomly chosen.
- You will receive \$10 if your answer is statistically correct.

Participants who made the recordings were informed they had a chance to win a bonus of \$10 if you selected the statistically correct answer. They were also informed that you had a chance to win a bonus of \$10 if you selected the statistically correct answer.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

1. In this study, you will get information about another respondent's answer to the task. Which one of the following statements is true?
 - I will listen to a voice message from another respondent about the task.
 - I will see what another respondent answered on the task.

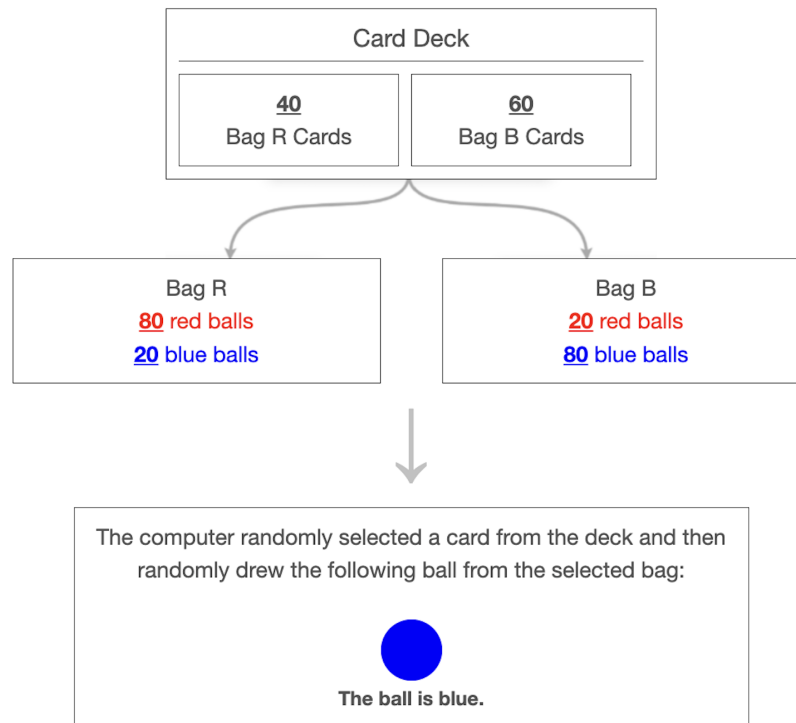
- I might listen to a voice message from another respondent about the task, or see what another respondent answered on the task.
2. Which one of the following statements is true?
- On a page with a voice recording, the recording only starts playing once I click “Play”.
 - On a page with a voice recording, the recording starts playing automatically when I enter, so I should read the question and pay attention to the recording.
3. Which one of the following statements is true?
- The answers I choose have no effect on my expected bonus.
 - I maximize my expected bonus by selecting my best answer to each question.
4. Which one of the following statements is true?
- I will be asked whether Bag R or Bag B is more likely to have been selected.
 - I will be asked whether the ball is more likely to be blue or red.
5. Which one of the following statements is true?
- If the computer draws a card with “Bag R” on it from the deck of cards, it will draw a ball from Bag R and show it to me.
 - If the computer draws a card with “Bag R” on it from the deck of cards, it will draw a ball from a randomly selected bag and show it to me.
6. Which one of the following statements is NOT true?
- Bags R and B always have the same proportion of red and blue balls.
 - Bag R always has at least as many red balls as blue balls.
 - Bag B always has at least as many blue balls as red balls.

Main Part

On the next page, the main task will be displayed.

[Prior Choice:] Provide your best answer

Please answer what you think is the correct answer to the question.



Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

- Bag R
- Bag B

How certain are you that your above answer is statistically correct?

[Slider from 0% (Very uncertain) to 100% (Completely certain)]

[Explanation Treatment:]

Now, you will listen to a recording of a voice message from a previous respondent who shares the explanation behind their answer to the exact same question that you just

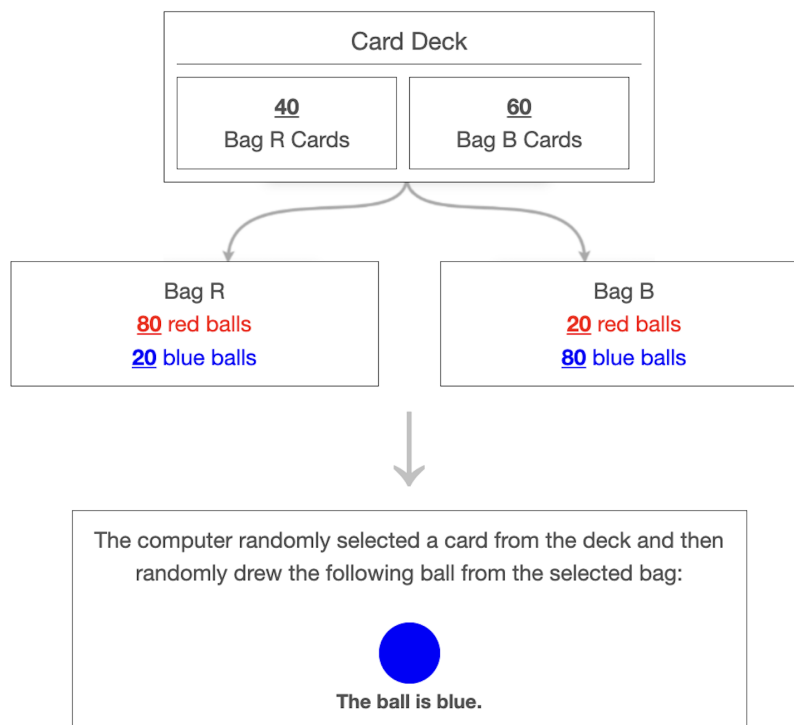
answered. The voice message will automatically start playing once you proceed to the next page.

Please listen closely to the recording.

You will be able to proceed once the recording has finished playing.

[PAGEBREAK]

Listen to the other respondent's answer



Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

Other person's answer:

[Answer of other respondent]

[Recording of other respondent, on auto-play]

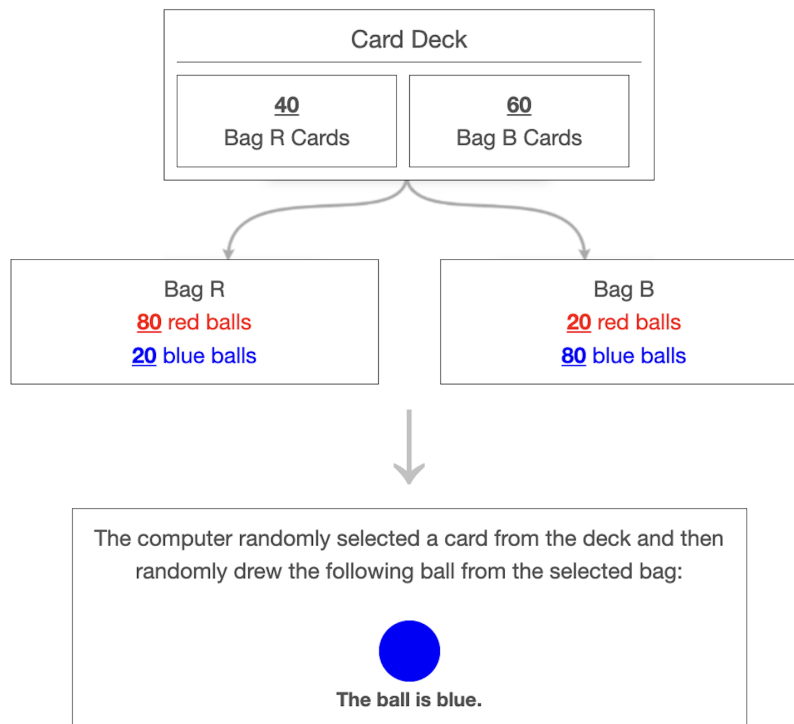
[Choice Only Treatment:]

Now, you will observe the answer from a previous respondent.

Please pay close attention to the other person's answer.

[PAGEBREAK]

Read the other respondent's answer



Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

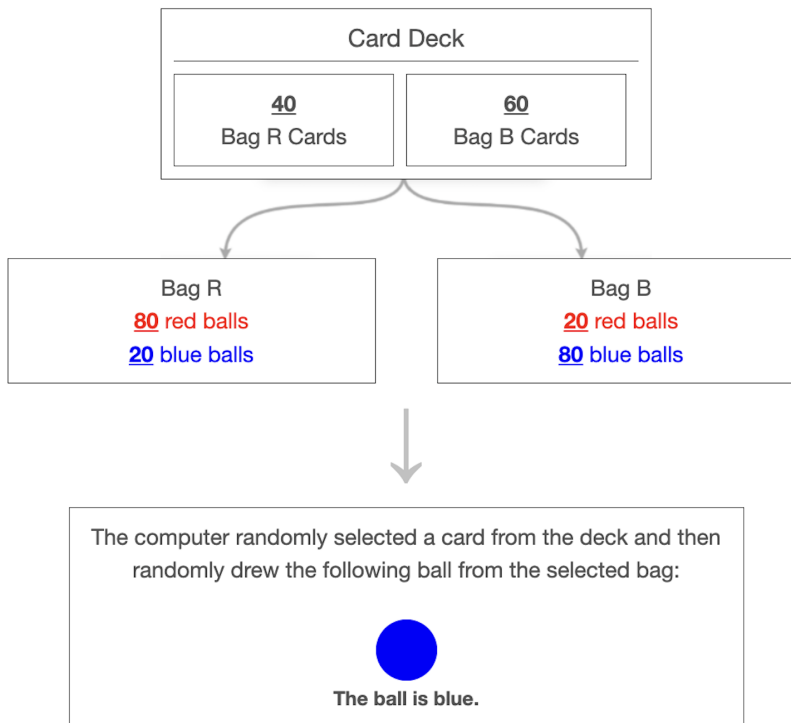
Other person's answer:

[Answer of other respondent]

[Posterior Choice:] Provide Your Best Answer

Your answers on this page may or may not be different from your previous response, given what you learned about the other participant's answer.

Please answer what you think is the correct answer to the question.



Given that this ball was drawn, which bag do you think is more likely to have been selected: Bag R or Bag B?

- Bag R
- Bag B

How certain are you that your above answer is statistically correct?

[Slider from 0% (Very uncertain) to 100% (Completely certain)]

Additional Questions

[Explanation Treatment:]

Your answer to the following question will not affect the other respondent's reward or bonus payment for this study, so please answer honestly. Did the other participant sound like they were using a computer-generated voice or reading from a script, for example one written by an AI assistant?

- Yes
- No

Your answer to the following question will not affect your reward or bonus payment for this study, so please answer honestly.

Did you search the answer to the question or use external tools, like an AI assistant, before providing your own answer?

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]

10 Argument annotation experiment

Welcome

In this survey, you will be asked to analyze explanations given by respondents in a previous survey. This study is designed for computer (PC or Mac) users only (desktop, laptop, etc.). If you are accessing this study on a smartphone, a tablet or any other non-PC device, please switch to PC and enter the study again, or return the submission on Prolific.

Please also complete the captcha below.

[Captcha]

Attention Check

This question is an attention check. To be eligible to participate in this study, we simply ask you to record a voice message that lasts at least 20 seconds on your opinion about daylight savings in the United States. Whether you are in favor or against daylight savings does not affect your eligibility to participate in this study. However, we ask that you record a message that lasts at least 20 seconds.

[Recording box, activated manually]

General Instructions

This study will take approximately 20 minutes to complete. You will earn a reward of \$5 for completing the survey. To complete the study, you will need to read all instructions carefully and correctly answer the comprehension questions.

Your Task

In a previous survey, participants answered a financial reasoning question. They recorded an oral explanation intended for a later participant, who would listen to it before answering the same question. These explanations were then transcribed and edited.

The most frequently used arguments in these explanations were subsequently identified. Your task is to determine which arguments appear in a set of 10 explanations.

For each of the 10 explanations, you will see a list of possible arguments. Select every argument that appears in the explanation — this may be several, one, or none. An explanation may contain multiple distinct arguments, a single argument, or none of the listed arguments at all.

You should identify arguments based on the reasoning the participant uses to justify their answer, not on their choice of words. Select an argument if the explanation expresses that reasoning, even if it uses completely different words. Do not select an argument merely because the explanation contains similar-sounding words or phrases; shared vocabulary without shared reasoning is not a match. When in doubt, ask: is the participant actually making this argument, or does it just sound like they might be? If the answer is the latter, do not select it.

For example, suppose a possible argument is that “Stocks are risky because their prices fluctuate unpredictably.” A respondent saying “you never know where the market will go day-to-day, so putting money in feels like gambling” does express this argument (different words, same reasoning). A respondent saying “stocks are risky — I wouldn’t touch them” does not (mentions risk but doesn’t give the fluctuation/unpredictability reasoning).

Bonus Payment

You are eligible for a bonus of up to \$10. Your chances of receiving the bonus are higher when your answers align with those of other participants.

More precisely, at the end of the survey:

- One of the 10 explanations you evaluated will be selected at random.
- One of the arguments you assessed for that explanation will be selected at random.
- You will then enter a lottery. Your probability of winning the \$10 bonus equals the share of other participants whose answer matches yours.

For example:

- If 70% of other participants selected an argument and you also selected it, your probability of winning is 70%.
- If 70% selected it and you did not select it, your probability of winning is 30%.

Comprehension Questions

Please answer the comprehension questions below. Note that if you fail them twice in a row, you will not be eligible for the completion payment.

In this study, you will read 10 explanations. Which one of the following statements is true?

- Each explanation will be about a different question.
- Each explanation will be about the same question about climate change.
- Each explanation will be about the same question about financial reasoning.

Which one of the following statements is true?

- I should always select exactly one argument.
- I should select every argument that appears in the explanation — this may be several, one, or none.

Which one of the following statements is true?

- If an explanation contains the same words as a specific argument, I should definitely mark the argument as being present in the explanation.
- I should only mark an argument as present if its logical substance is expressed, even if the explanation uses different words.

Which one of the following statements is true?

- The answers I choose have no effect on my expected bonus.
- I maximize my expected bonus by identifying arguments in a way that matches other participants' responses.

Main Part

On the following page, the main part of the survey starts and you will see your first question.

Question [1/10]

The question will be displayed on the next page.

[PAGEBREAK]

Question:

[Question text]

Other person's answer:

[Answer of other respondent]

Other person's explanation:

[Edited transcript of explanation]

Which arguments did the person above use? Select every argument that appears in the explanation — this may be several, one, or none.

[List of possible arguments, displayed as multi-selectable checkboxes]

Additional Questions

Did you look up any answers on the internet or use external tools? Your response to this question will not affect your payment. Please answer truthfully.

- Yes
- No

[Followed, on a separate page, by elicitation of sex, age, ethnicity, education, employment and political affiliation.]